

Nama : Rengganis Tantri Pramudita

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Jurnal Modul 6

Link GitHub: https://github.com/rengganistan/KPL_Rengganis-Tantri-Pramudita_2311104065_SISE-07-02.git

1. MENAMBAHKAN KODE AWAL SAYATUBEVIDEO

- SayaTubeVideo.cs

```
class SayaTubeVideo
{
    private static Random rand = new Random();
    private int id;
    private string title;
    private int playCount;

    public SayaTubeVideo(string title)
    {
        if (string.IsNullOrEmpty(title) || title.Length > 200)
            throw new ArgumentException("Title cannot be empty or exceed 200 characters");

        this.id = rand.Next(10000, 99999);
        this.title = title;
        this.playCount = 0;
    }

    public void IncreasePlayCount(int count)
    {
        if (count < 0 || count > 10000000)
            throw new ArgumentOutOfRangeException("Count must be between 0 and 10,000,000");

        this.playCount += count;
    }

    public void PrintVideoDetails()
    {
        Console.WriteLine($"ID: {id}");
        Console.WriteLine($"Title: {title}");
        Console.WriteLine($"Play Count: {playCount}");
        Console.WriteLine("-----");
    }
}
```

```

    }

    public int GetPlayCount()
    {
        return playCount;
    }

    public string GetTitle()
    {
        return title;
    }
}

```

- SayaTubeUser.cs

```

class SayaTubeUser
{
    private static Random rand = new Random();
    private int id;
    private List<SayaTubeVideo> uploadedVideos;
    public string Username { get; private set; }

    public SayaTubeUser(string username)
    {
        if (string.IsNullOrEmpty(username) || username.Length > 100)
            throw new ArgumentException("Username cannot be empty or exceed 100 characters");

        this.id = rand.Next(10000, 99999);
        this.Username = username;
        this.uploadedVideos = new List<SayaTubeVideo>();
    }

    public void AddVideo(SayaTubeVideo video)
    {
        if (video == null)
            throw new ArgumentNullException("Video cannot be null");

        if (uploadedVideos.Count >= 10)
            throw new InvalidOperationException("Maximum of 10 videos can be uploaded.");

        uploadedVideos.Add(video);
    }

    public int GetTotalVideoPlayCount()
    {
        int total = 0;
        foreach (var video in uploadedVideos)
        {
            total += video.GetPlayCount();
        }
        return total;
    }
}

```

```

    }

    public void PrintAllVideoPlaycount()
    {
        Console.WriteLine($"User: {Username}");
        Console.WriteLine("Daftar Video yang Diupload:");
        Console.WriteLine("-----");
        for (int i = 0; i < uploadedVideos.Count; i++)
        {
            Console.WriteLine($"Video {i + 1} judul: {uploadedVideos[i].GetTitle()}");
        }
        Console.WriteLine("-----");
    }
}

```

- Program.cs

```

class Program
{
    static void Main()
    {
        SayaTubeUser user = new SayaTubeUser("Rengganis");

        string[] filmTitles = {
            "Inception", "Interstellar", "The Dark Knight", "Parasite",
            "Avengers: Endgame", "Forrest Gump", "The Shawshank Redemption",
            "Titanic", "Spirited Away", "The Matrix"
        };

        foreach (string title in filmTitles)
        {
            SayaTubeVideo video = new SayaTubeVideo($" {title} oleh Rengganis");
            user.AddVideo(video);
        }

        user.PrintAllVideoPlaycount();
    }
}

```

Hasil run:

```
Microsoft Visual Studio Debug Console
Use the dropdown to view and switch to other projects this file may belong to.

User: Rengganis
Daftar Video yang Diupload:
-----
Video 1 judul: Inception oleh Rengganis
Video 2 judul: Interstellar oleh Rengganis
Video 3 judul: The Dark Knight oleh Rengganis
Video 4 judul: Parasite oleh Rengganis
Video 5 judul: Avengers: Endgame oleh Rengganis
Video 6 judul: Forrest Gump oleh Rengganis
Video 7 judul: The Shawshank Redemption oleh Rengganis
Video 8 judul: Titanic oleh Rengganis
Video 9 judul: Spirited Away oleh Rengganis
Video 10 judul: The Matrix oleh Rengganis
-----

C:\Users\asus-\source\repos\KPL_Rengganis-Tantri-Pramudita_2311104065_S1SE-gramming\jurnal\modul6_2311104065\modul6_2311104065\bin\Debug\net8.0\modul6ode 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Environment->Close console when debugging stops.
Press any key to close this window . . .
```

Penjelasan:

Hasil run menunjukkan bahwa pengguna "Rengganis" berhasil mengunggah daftar 10 video dengan judul film terkenal yang sesuai dengan format yang telah ditentukan. Namun, karena postcondition membatasi jumlah video yang ditampilkan maksimal 8, implementasi kode saat ini masih mencetak semua 10 video tanpa filter. Selain itu, program berjalan tanpa error atau crash, menunjukkan bahwa precondition dan exception handling berfungsi dengan baik untuk memastikan input valid serta mencegah kesalahan yang dapat menyebabkan program berhenti secara tiba-tiba.

2. MENAMBAHKAN IMPLEMENTASI DESIGN BY CONTRACT

- SayaTubeVideo.cs

```
class SayaTubeVideo
{
    private int id;
    private string title;
    private int playCount;

    public SayaTubeVideo(string title)
```

```

{
    if (string.IsNullOrEmpty(title) || title.Length > 200)
        throw new ArgumentException("Title cannot be empty or exceed 200 characters");

    Random rand = new Random();
    this.id = rand.Next(10000, 99999);
    this.title = title;
    this.playCount = 0;
}

public void IncreasePlayCount(int count)
{
    if (count < 0 || count > 25000000)
        throw new ArgumentOutOfRangeException("Count must be between 0 and 25,000,000");

    try
    {
        checked
        {
            this.playCount += count;
        }
    }
    catch (OverflowException)
    {
        Console.WriteLine("Error: Play count exceeded the maximum limit!");
    }
}

public void PrintVideoDetails()
{
    Console.WriteLine($"ID: {id}");
    Console.WriteLine($"Title: {title}");
    Console.WriteLine($"Play Count: {playCount}");
}

public int GetPlayCount()
{
    return playCount;
}

```

```
public string GetTitle()
{
    return title;
}
}
```

- SayaTubeUser.cs

```
class SayaTubeUser
{
    private int id;
    private List<SayaTubeVideo> uploadedVideos;
    public string Username { get; private set; }

    public SayaTubeUser(string username)
    {
        if (string.IsNullOrEmpty(username) || username.Length > 100)
            throw new ArgumentException("Username cannot be empty or exceed 100 characters");

        this.id = new Random().Next(10000, 99999);
        this.Username = username;
        this.uploadedVideos = new List<SayaTubeVideo>();
    }

    public void AddVideo(SayaTubeVideo video)
    {
        if (video == null)
            throw new ArgumentNullException("Video cannot be null");

        if (video.GetPlayCount() >= int.MaxValue)
            throw new OverflowException("Video play count is too high!");

        uploadedVideos.Add(video);
    }

    public int GetTotalVideoPlayCount()
    {

```

```

        int total = 0;
        foreach (var video in uploadedVideos)
        {
            total += video.GetPlayCount();
        }
        return total;
    }

    public void PrintAllVideoPlaycount()
    {
        Console.WriteLine($"User: {Username}");
        for (int i = 0; i < Math.Min(uploadedVideos.Count, 8); i++)
        {
            Console.WriteLine($"Video {i + 1} judul: {uploadedVideos[i].GetTitle()}");
        }
    }
}

```

- Program.cs

```

class Program
{
    static void Main()
    {
        SayaTubeUser user = new SayaTubeUser("Rengganis");

        string[] filmTitles = { "Inception", "Interstellar", "The Dark Knight", "Parasite", "Avengers:
Endgame", "Forrest Gump", "The Shawshank Redemption", "Titanic", "Spirited Away", "The
Matrix" };

        for (int i = 0; i < filmTitles.Length; i++)
        {
            SayaTubeVideo video = new SayaTubeVideo($" {filmTitles[i]} oleh Rengganis");
            user.AddVideo(video);
        }

        user.PrintAllVideoPlaycount();

        // Menguji Precondition & Exception
        try

```

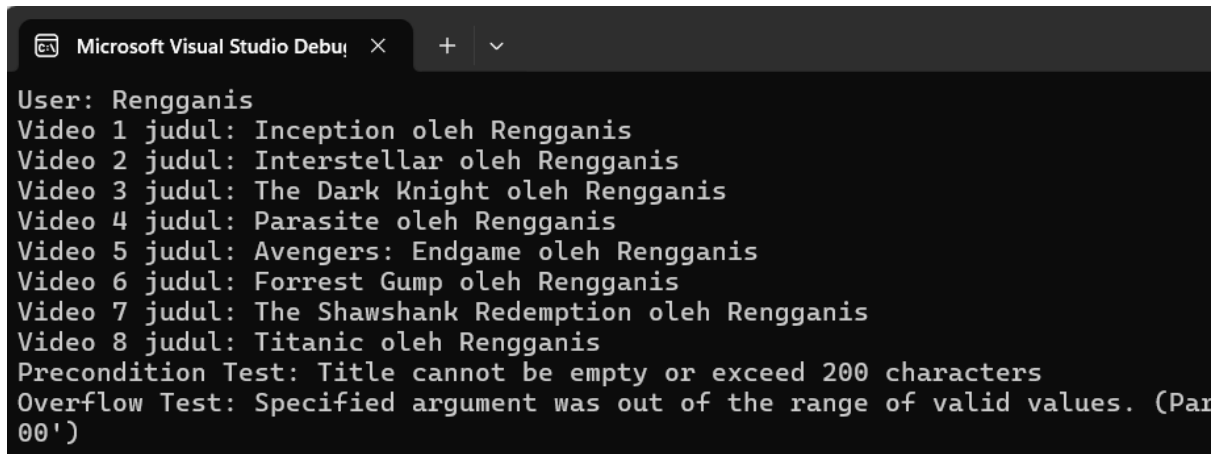
```

    {
        SayaTubeVideo invalidVideo = new SayaTubeVideo(""); // Ini akan menyebabkan
exception
    }
    catch (Exception e)
    {
        Console.WriteLine("Precondition Test: " + e.Message);
    }

    try
    {
        SayaTubeVideo normalVideo = new SayaTubeVideo("Test Video");
        normalVideo.IncreasePlayCount(int.MaxValue); // Akan memicu overflow exception
    }
    catch (Exception e)
    {
        Console.WriteLine("Overflow Test: " + e.Message);
    }
}
}

```

Hasil run:



```

Microsoft Visual Studio Debug Console
User: Rengganis
Video 1 judul: Inception oleh Rengganis
Video 2 judul: Interstellar oleh Rengganis
Video 3 judul: The Dark Knight oleh Rengganis
Video 4 judul: Parasite oleh Rengganis
Video 5 judul: Avengers: Endgame oleh Rengganis
Video 6 judul: Forrest Gump oleh Rengganis
Video 7 judul: The Shawshank Redemption oleh Rengganis
Video 8 judul: Titanic oleh Rengganis
Precondition Test: Title cannot be empty or exceed 200 characters
Overflow Test: Specified argument was out of the range of valid values. (Parameter 'value')

```

Penjelasan:

Saat program dijalankan, output yang dihasilkan menampilkan daftar video yang telah ditambahkan oleh pengguna "Rengganis", dengan maksimal 8 judul video sesuai dengan postcondition yang diterapkan. Jika terdapat input yang melanggar precondition, seperti judul video kosong atau play count melebihi 25.000.000, program akan menampilkan pesan error tanpa langsung berhenti. Selain itu,

mekanisme exception handling juga mencegah terjadinya overflow pada play count, sehingga program tetap berjalan dengan aman meskipun ada percobaan untuk melebihi batas maksimum integer.